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Outlier Detection

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Outliers exist in almost every real life data

Malicious activity

Insurance fraud, credit card fraud, telecom fraud, cyber intrusion, terrorist activity

Instrumentation error

Defects in components of machines or wear and tear in equipment

Change in the environment

Climate change, new buying pattern among consumers, mutation in genes

Human error

Automobile accident, data reporting error

Definitions

- ***Outliers are defined*** typically based on the following assumptions:

  (1) Outliers are different from the norm with respect to their features.

  (2) outliers are rare in a dataset compared to normal instances.

- In a better way:
  - Data points that deviate significantly from the rest.

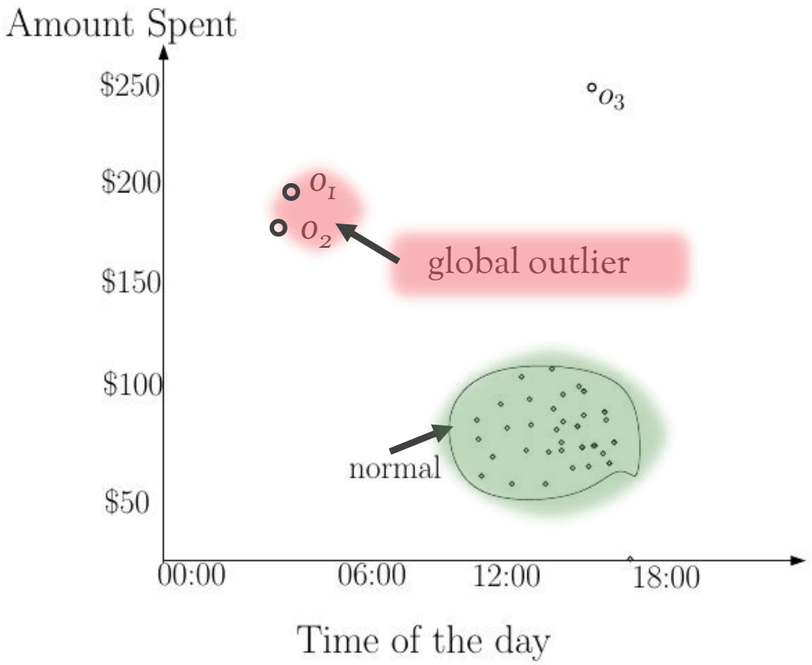
- Outliers can have a significant impact on the statistical analysis of a dataset. They can skew the results of statistical tests and lead to incorrect conclusions.

- Therefore, it's essential to identify and handle outliers appropriately.

Type of Outlier

Type I - Global Outlier

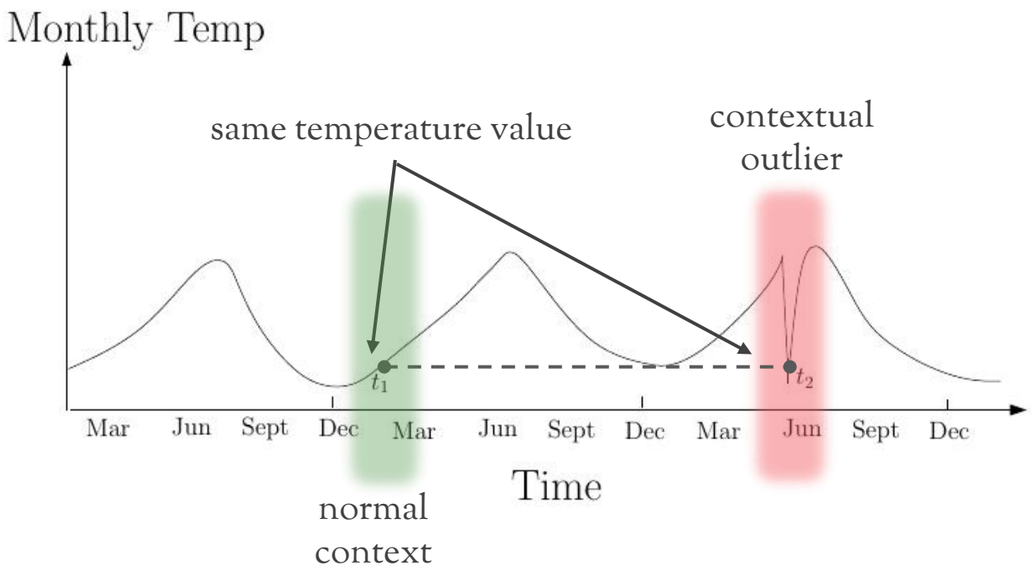
- A data instance is an outlier due to its attribute values which are inconsistent with values taken by normal instances.
- Ex: Transactions o1 and o2 are fraudulent transactions which are outliers because they occur at an abnormal time and the amount is abnormally large.



Type of Outlier

Type II - Contextual Outliers (conditional outliers)

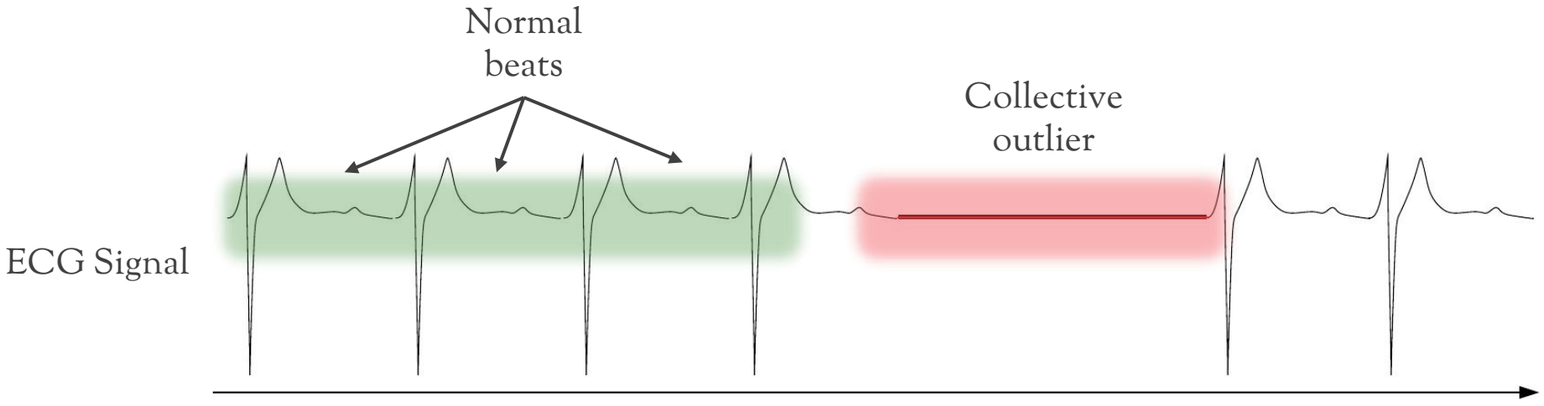
- These outliers are caused due to the occurrence of an individual data instance in a specific context in the given data.
- Ex: the temperature at time t1 is same as that at time t2 but occurs in a different context and hence is not considered as an outlier.



Type of Outlier

Type III - Collective Outlier

- These outliers occur because a subset of data instances are outlying with respect to the entire data set.
- Outlier in an human electrocardiogram output, even if the low value in the flatline also occurs in normal regions of the sequence.



Outlier vs. Anomaly vs. Novel Pattern vs. Noise

- ***Outliers*** tend to emphasize statistical rarity and deviation, and whether they are generated by a different mechanism is not straightforwardly addressed.
- In contrast, ***anomalies*** suggest a different underlying generative mechanism.
- ***Noise*** is a random, meaningless variations that obscure true data patterns; typically removed during data preprocessing.
- While, the ***novel patterns*** are typically incorporated with the normal model after getting detected.

Concept	Definition	Example	Requires Investigation?
Outlier	*Extreme values in data*	A house priced at $10M in a dataset of $100K–$500K houses	Sometimes (may be noise)
Anomaly	*A deviation from expected behavior*	A credit card used in another country for a large purchase	Yes (could indicate fraud)
Novel Pattern	*A new but valid pattern in data*	A surge in purchases on Black Friday	No (may be a trend)
Noise	*Random, meaningless variations in data*	Sensor malfunction causing fluctuating temperature readings	No (should be removed)

Taxonomy of Outlier Detection Methods

Graphical Approach

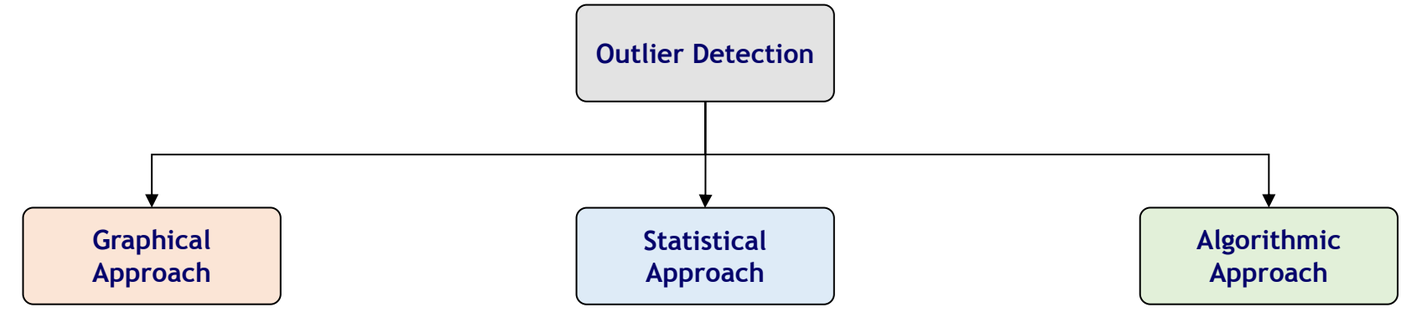
- detect outliers through data visualizations.

Statistical Approach

- detect outliers through statistical analysis and probability distributions.

Algorithmic Approach

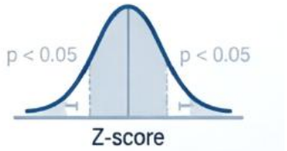
- detect outliers through Machine Learning models.



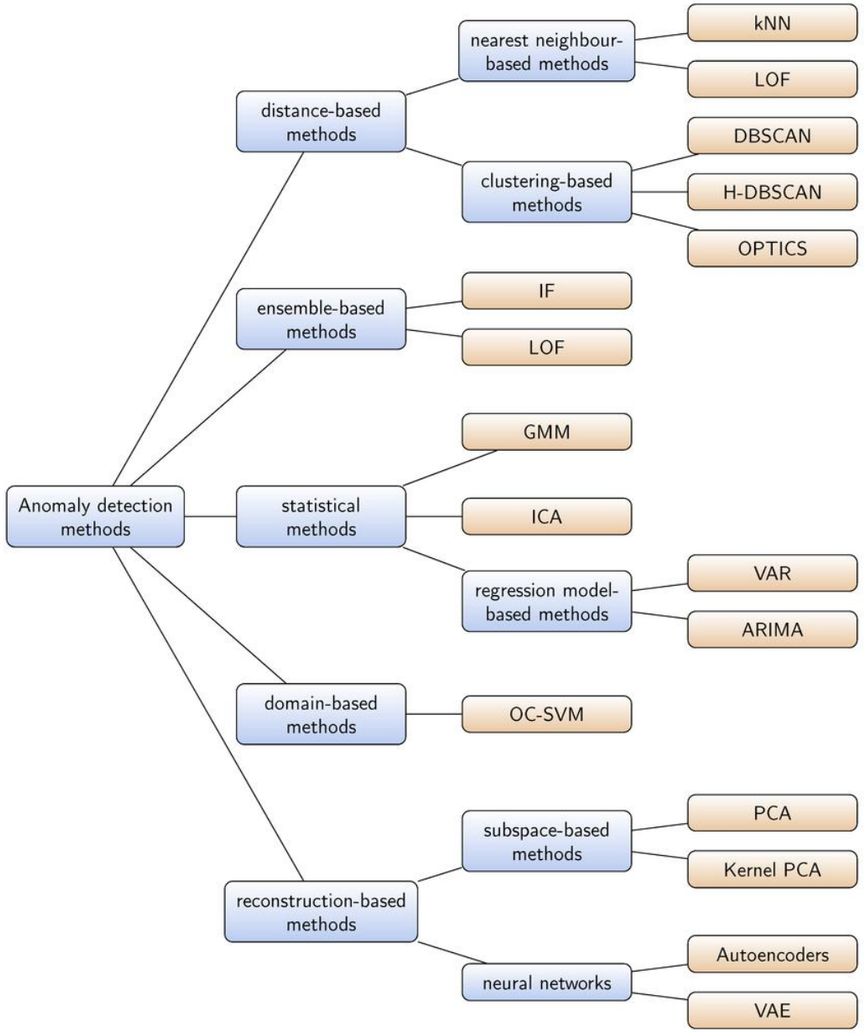
- Scatter Plot
- Box Plot
- Histograms
- Q-Q Plots
- Violin Plot
- Regression Plot
- Heatmaps
- Hexbin plot

- Z-Score
- Interquartile Range (IQR)
- Tukey's Fences
- Grubbs' Test
- Dixon's Q Test

- Isolation Forest
- Local Outlier Factor (LOF)
- One-Class SVM
- Autoencoders
- DBSCAN



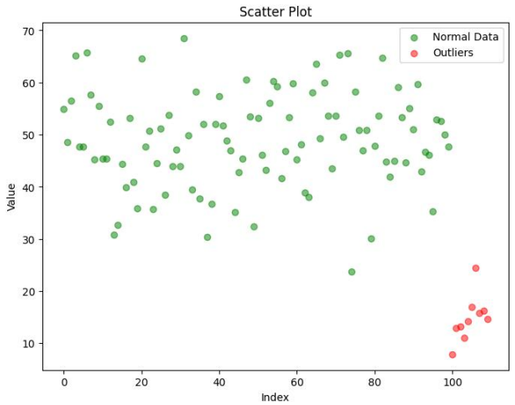
Other classification

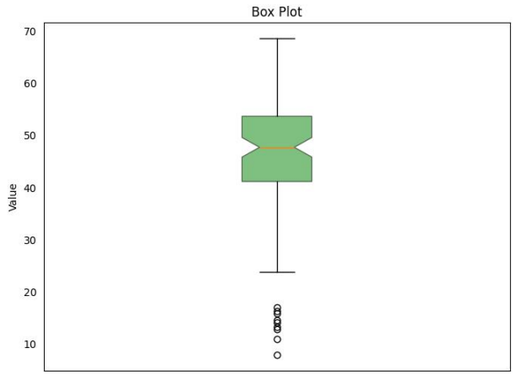


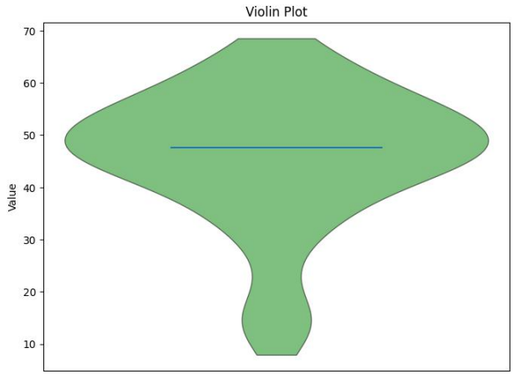
Graphical Approach

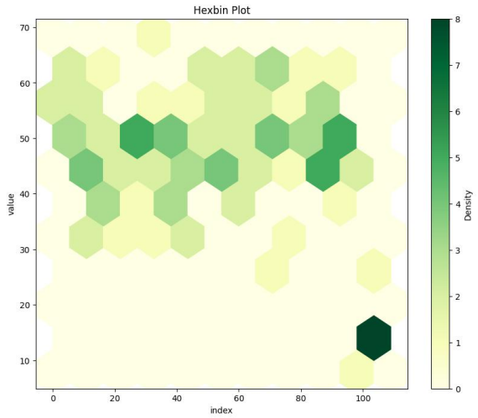
The Graphical Approach for outlier detection leverages the human brain's remarkable ability to discern patterns.

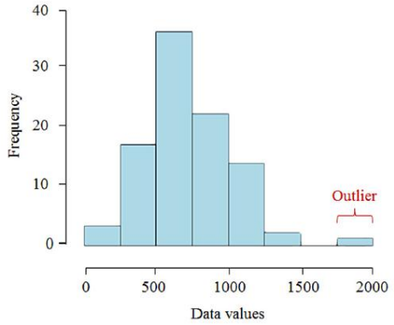
It makes use of visualization tools like Scatter Plots, Box Plots, and Heatmaps to provide a narrative of the data and allow Data Scientists to spot irregularities in the pattern.

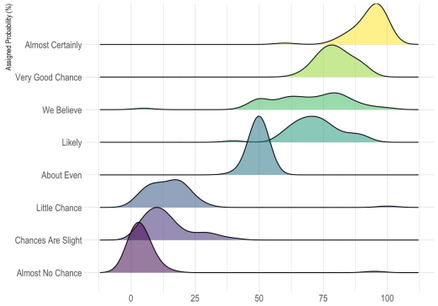


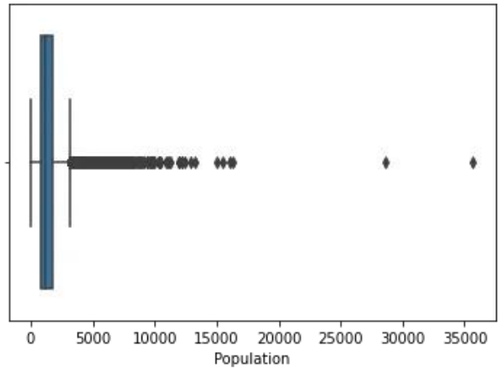










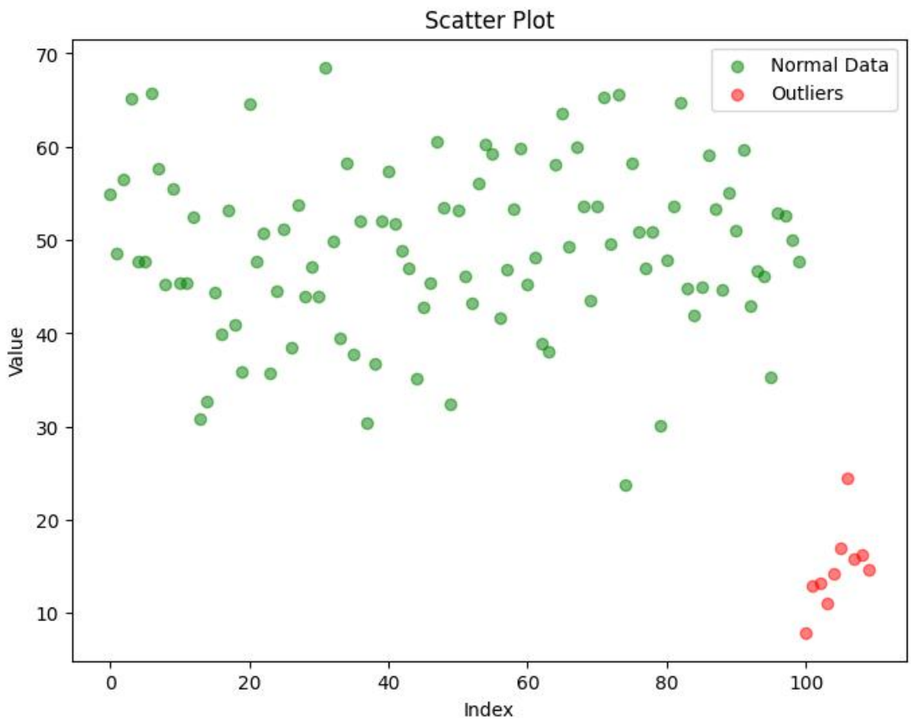


Graphical Approach - Scatter Plot

In a Scatter Plot, outliers will appear as points significantly deviating from the main cluster.

```python
# Generate synthetic data with outliers
np.random.seed(42)
normal_data = np.random.normal(loc=50, scale=10, size=100)
outliers = np.random.normal(loc=15, scale=5, size=10)
```

```python
# Combine normal data with outliers
data = np.concatenate((normal_data, outliers))
# Create index ranges for plotting
range_1 = range(len(normal_data))
range_2 = range(len(normal_data), len(data))
# Visualize data using a scatter plot
fig, ax = plt.subplots(figsize=(8, 6))
ax.scatter(range_1, normal_data, color="green", alpha=.5, label='Normal Data')
ax.scatter(range_2, outliers, color="red", alpha=.5, label='Outliers')
```

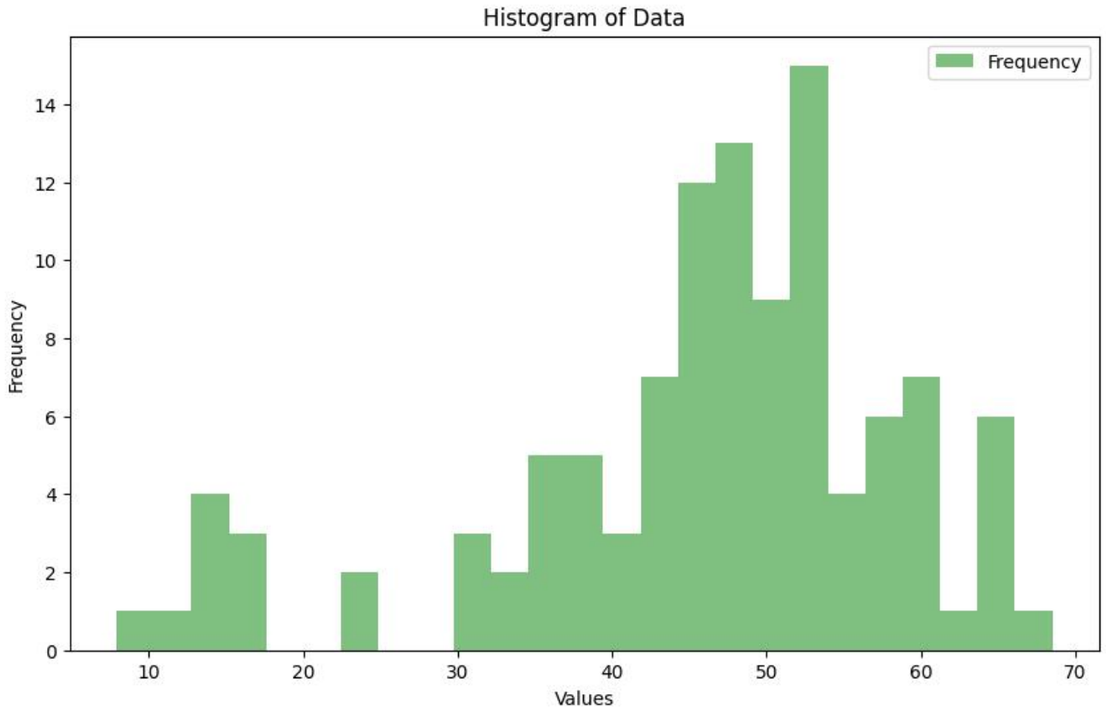


Graphical Approach - Histogram Plot

Histograms serve as effective tools for detecting outliers through visual inspection.

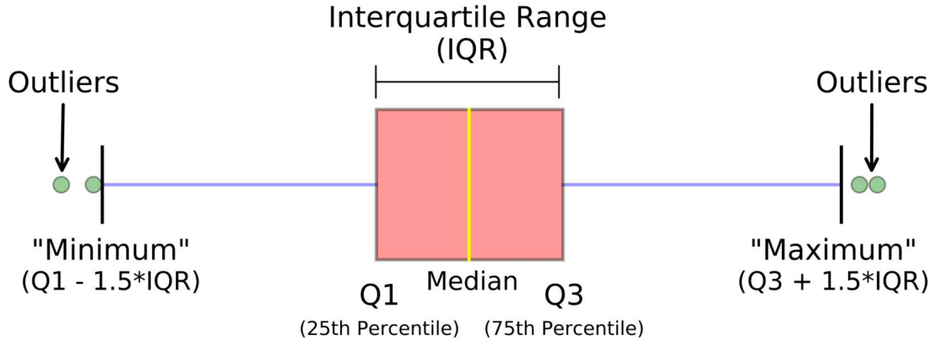
Outliers often appear in the tails or sparse bins of the histogram.

```
plt.hist(data, bins=25, alpha=0.5, color="green", label='Frequency')
```

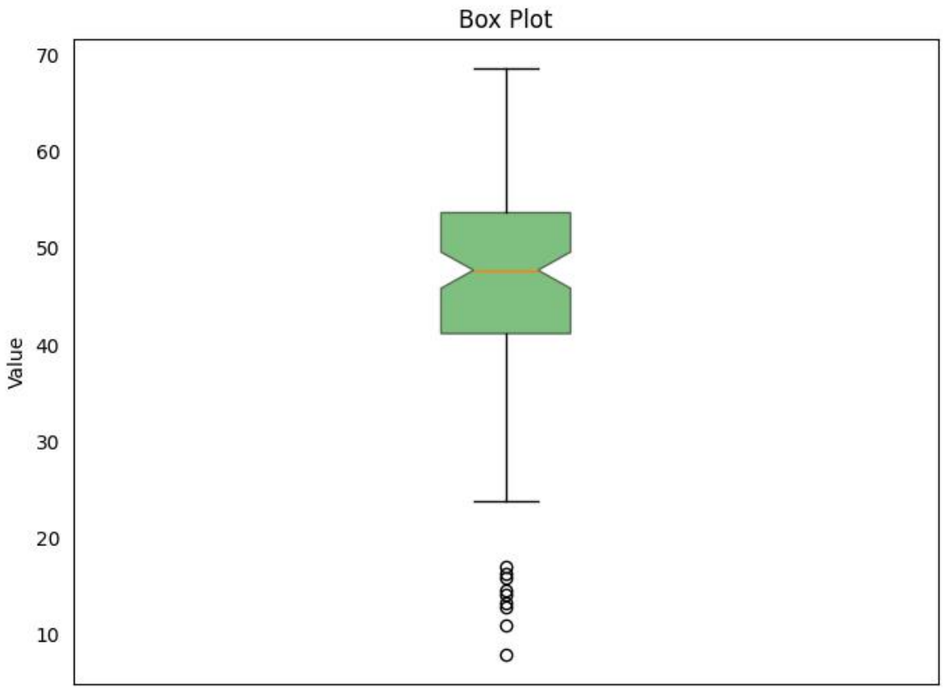


Graphical Approach - Box Plot

Using the same data, we can display a Box Plot, in which outliers appear as points outside the "whiskers" of the Box Plot.




```python
# Visualize data using a box plot
fig, ax = plt.subplots(figsize=(8, 6))
b_plot = ax.boxplot(data, vert=True, patch_artist=True, notch=True)
ax.set_ylabel('Value')
ax.set_title('Box Plot')
# Color the box
for box in b_plot['boxes']:
    box.set_facecolor("green")
    box.set_alpha(.5)
```

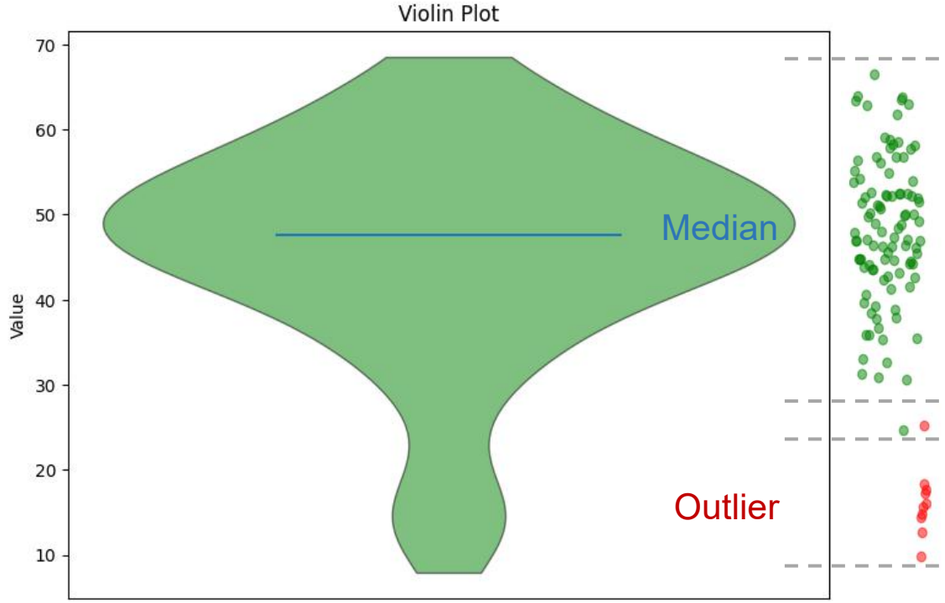


Graphical Approach - Violin Plot

Violin Plots don't show only the distribution of the data but also display its Probability Density.

Outliers are visible as a thin section extending beyond the bulk of the data.

```python
# Visualize data using a violin plot
fig, ax = plt.subplots(figsize=(8, 6))
v_plot = ax.violinplot(data, vert=True, showmedians=True, showextrema=False)
ax.set_ylabel('Value')
ax.set_title('Violin Plot')
# Color the violin
for pc in v_plot['bodies']:
    pc.set_facecolor('green')
    pc.set_edgecolor('black')
    pc.set_alpha(.5)
```



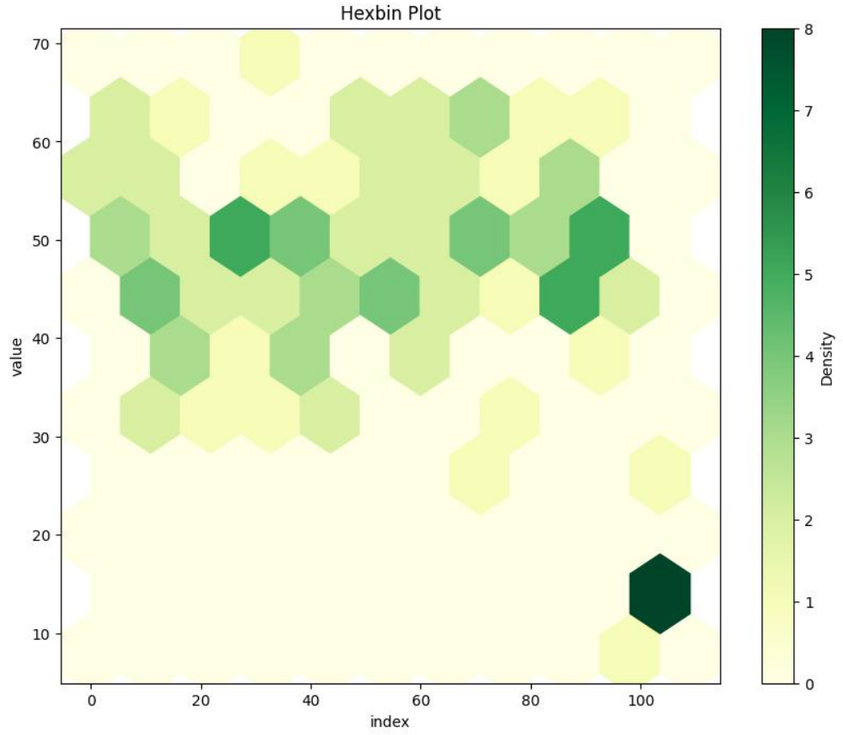
Graphical Approach - Hexbin plot

Hexbin plot is useful for outlier detection when analyzing high-density areas and spotting points in **low-density regions** and **isolated bins**.

The main idea is:

- **Dense clusters** represent normal values.
- **Sparse or isolated bins** may indicate outliers.
- **Color intensity** helps visualize concentration.

```
# Visualize data using a hexbin plot
hb = ax.hexbin(x_data, y_data, gridsize=10, cmap='YlGn')
```



Statistical Approach

While the graphical approach is indeed of more immediate understanding, it comes with limitations. The main issue is they provide just ***qualitative*** but ***not quantitative*** results.

For consistent analysis, we must rely on mathematical rigor and statistical metrics.

Parametric vs. ***Non-Parametric*** Method:

- Parametric methods assume that ***data follow a specific distribution***, often a ***normal distribution***.
- Non-parametric methods, also known as ***distribution-free methods***, ***do not*** assume a specific ***distribution for the data***.

Statistical Approach - Z-Score

It measures the number of standard deviations away the observation is from the mean value. For example, a z-score of 2 indicates that the observation is two standard deviation above the mean whereas z-score of -2 indicates the observation is two standard deviation below the mean.

To calculate the z-score, subtract the mean from the actual observation nd divide it by standard deviation. The further away an observation's Z-score is from zero, the more unusual it is. A standard cut-off value for finding outliers are Z-scores of +3 or -3 further from zero.

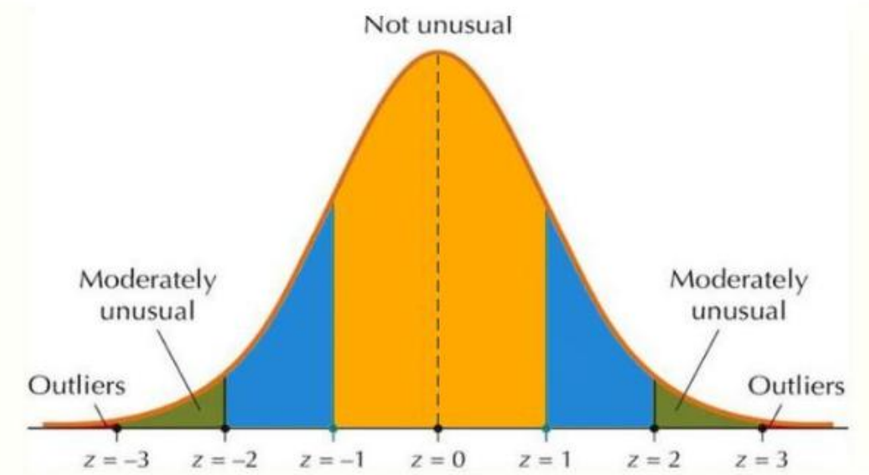
```python
#importing the libraries
import numpy as np
from scipy import stats
#calculate Z-score
z = np.abs(stats.zscore(df_California['Population']))
#general assumption of taking threshold as 3
threshold = 3
print(np.where(z>3))
```

Data point

Mean

$$z = \frac{(x - \mu)}{\sigma}$$

Standard deviation



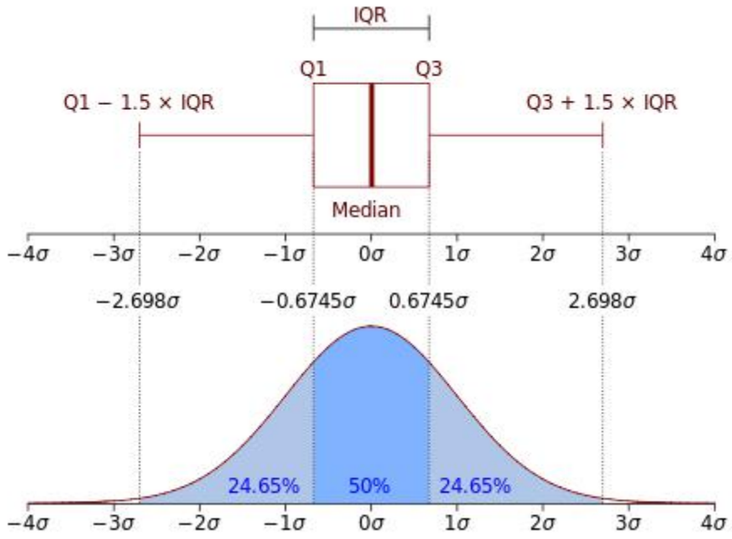
Statistical Approach - IQR Method - Non-parametric method

The ***Inter Quartile Range(IQR)*** is the middle 50% of the dataset. It's the range of values between the third quartile and the first quartile (Q3 — Q1).

- We can take the IQR, Q1, and Q3 values to calculate the following outlier fences for our dataset.
- IQR = Q3 — Q1
- Upper fence = Q1 — (1.5*IQR)
- Lower fence = Q3 + (1.5*IQR)

Data points outside the range [Q1–1.5 * IQR, Q3 + 1.5 * IQR] are often considered outliers.

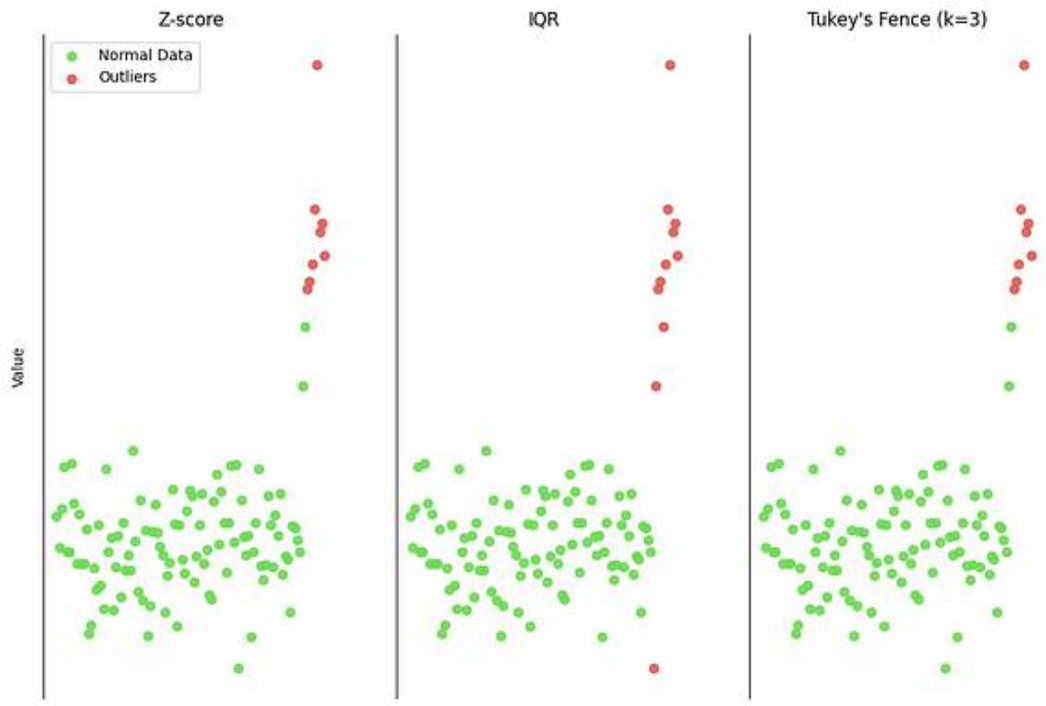
```python
# Calculate Q1 and Q3
Q1 = np.percentile(data, 25)
Q3 = np.percentile(data, 75)
# Calculate IQR
IQR = Q3 - Q1
# Set lower and upper bounds for outliers
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
# Identify outliers using IQR method
outliers = [value for value in data if value < lower_bound or value > upper_bound]
```



Statistical Approach - Tukey's Fence - Non-Parametric Method

Tukey's Fence is a non-parametric method for detecting outliers in univariate data, utilizing the interquartile range (IQR) to establish boundaries beyond which data points are considered anomalous.

It considers outliers all the data points that fall outside the range Q1-k*IQR and Q3+k*IQR, where k is a constant. Usually, k takes a value between 1.5 and 3.

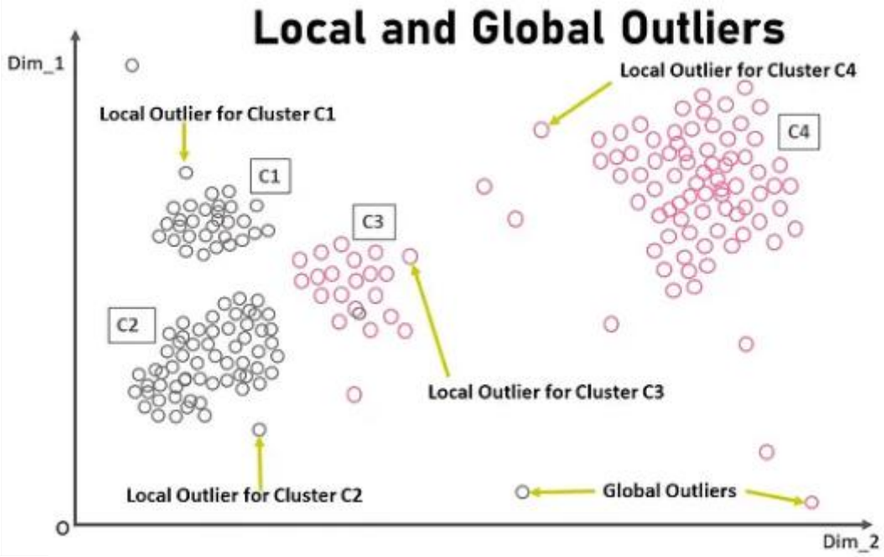


Algorithmic Approach - Local Outlier Factor (LOF)

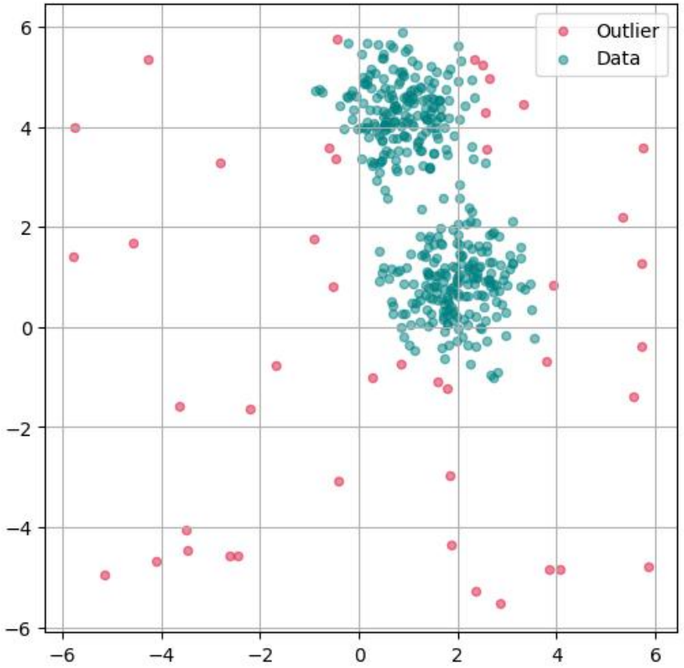
Local Outlier Factor (LOF) is particularly designed for identifying outliers in datasets where the density varies across different regions.

LOF relies on the concept of **Local Density Estimation**, where locality is given by the *k*-nearest neighbors of a point.

It compares the local density of a point to the densities of its neighbors. An outlier will have a substantially lower density than its neighbors.




```python
from sklearn.neighbors import LocalOutlierFactor

# Create the model
lof = LocalOutlierFactor(
    n_neighbors=20,
    contamination=0.1,
    metric_params=None,
    n_jobs=-1)

# Fit the model and classify outliers
outlier_classification = lof.fit_predict(X)
```

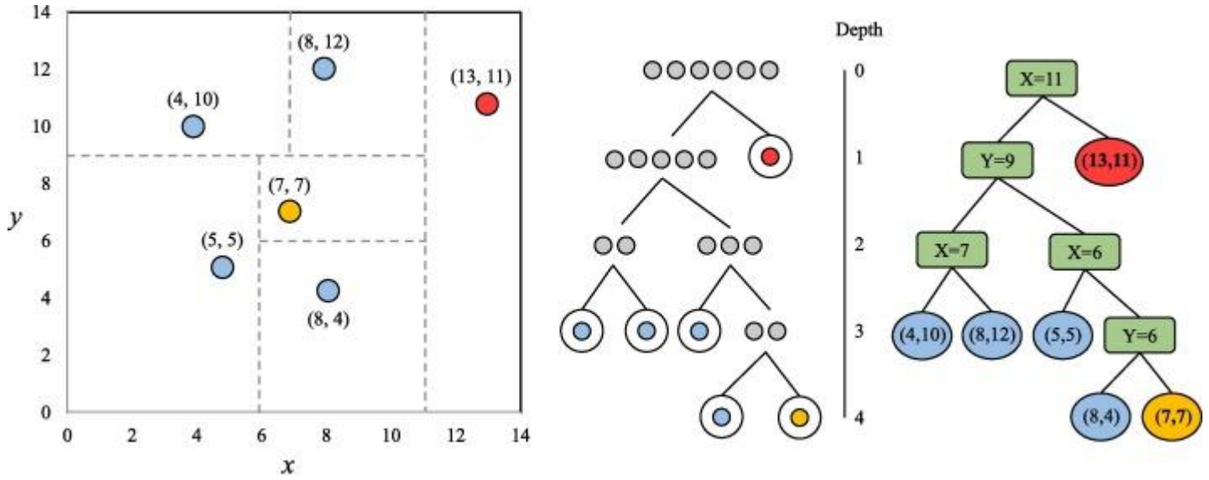


Algorithmic Approach - Isolation Forest

Isolation Forest (iForest) developed in 2008, it's particularly effective in datasets with a high dimensionality, works on the principle of isolating anomalies, instead of profiling normal data points.

It builds a set of trees, if the average number of splits required to isolate a given observation is low, that observation is considered a stronger candidate outlier.

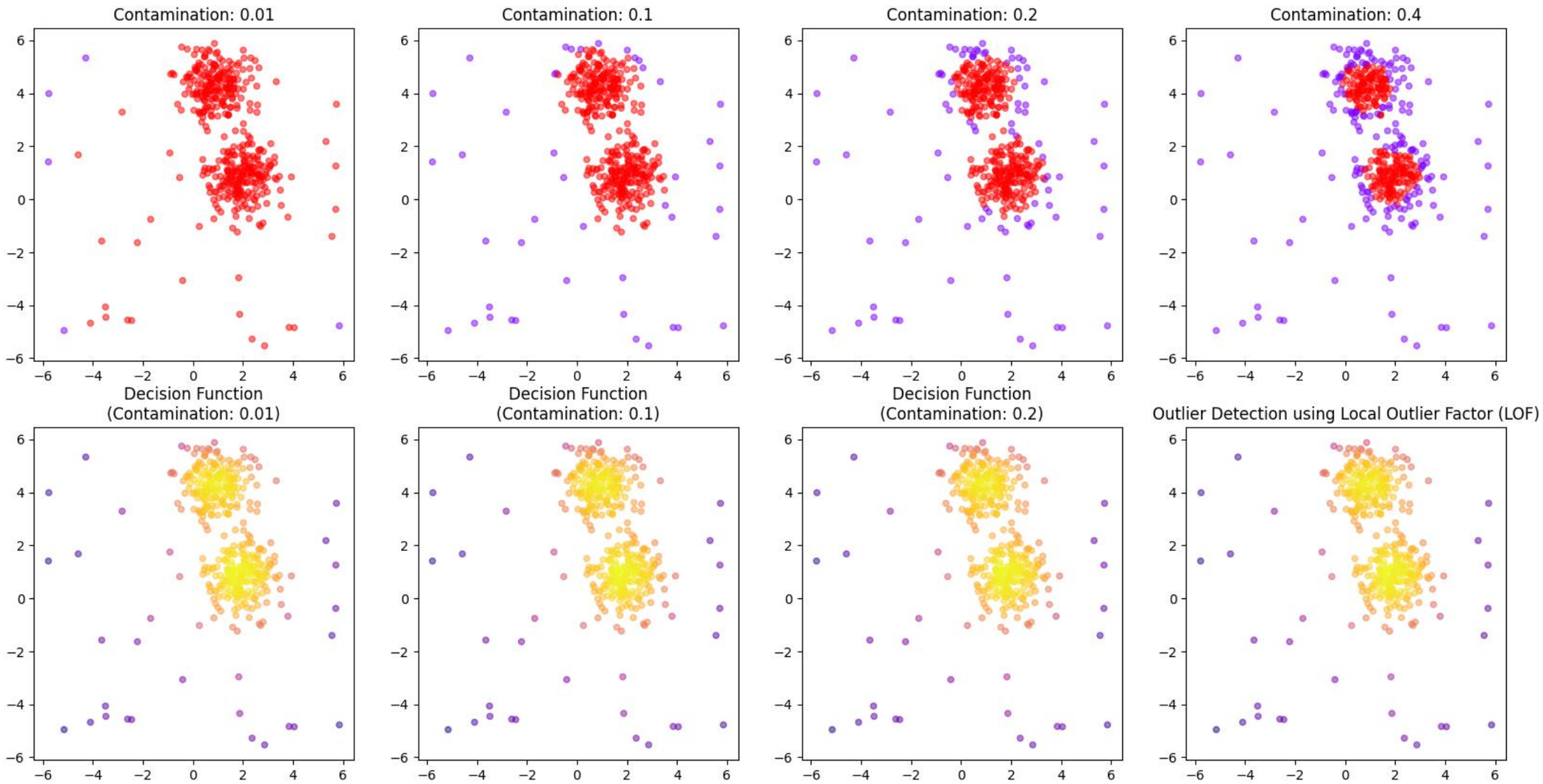
Like other tree-based models, Isolation Forest does not assume that the features are normally distributed or require them to be scaled.




```python
from sklearn.ensemble import IsolationForest

# Create the model
iso_forest = IsolationForest(
    n_estimators=100,
    contamination=0.075,
    n_jobs=-1,
    random_state=42,
    verbose=0)

# Fit the model
iso_forest.fit(X)

# Classify as outlier / no outlier
outlier_classification = iso_forest.predict(X)
```

`contamination` : The proportion of outliers in the data set.

Algorithmic Approach - Isolation Forest



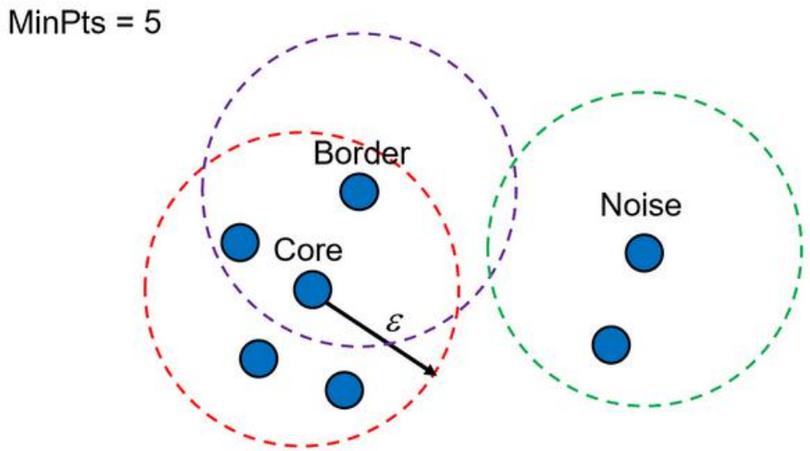
Algorithmic Approach - DBSCAN

Density-Based Spatial Clustering of Applications with Noise (DBSCAN), it clusters data points that are close together and identifies points that are far away from others as noise or outliers.

DBSCAN starts with a random point and checks how many other points fall within a certain distance, known as Epsilon (ε).

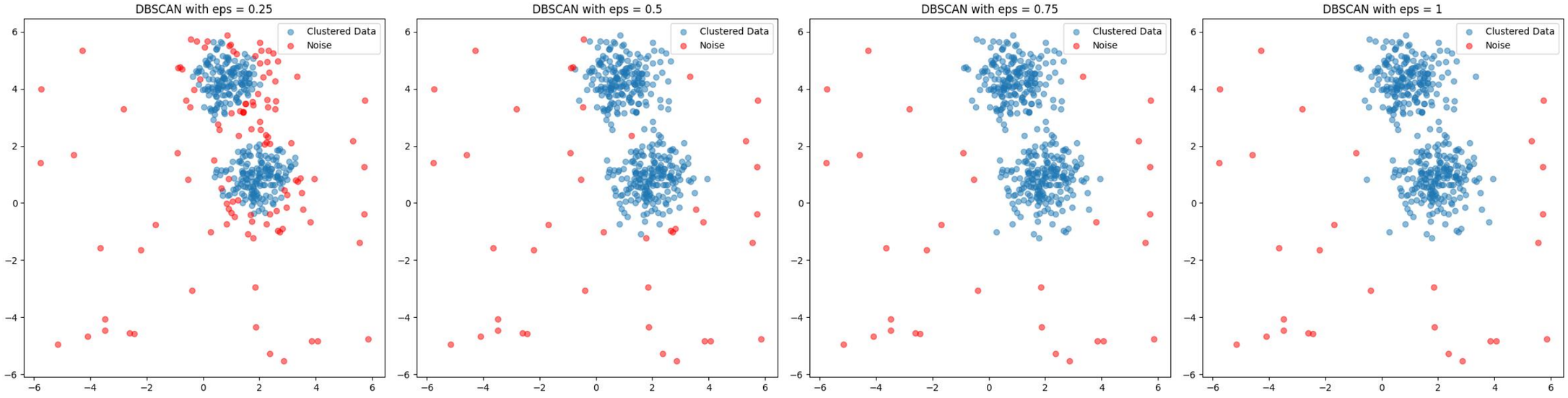
If there are enough points within this distance, more than a certain threshold called (MinPts) then that point is considered a “core point,” and a cluster begins to form.

And those lonely points that don’t belong to any cluster are labeled as “noise.”




```python
from sklearn.cluster import DBSCAN

# Create the model
dbscan = DBSCAN(eps=0.5, min_samples=5) # Adjust eps and min_samples as needed

# Fit the model
dbscan.fit(X)

# Get the labels, -1 represents noise/outliers
labels = dbscan.labels_
noise_mask = labels == -1
```

Algorithmic Approach - DBSCAN



References

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Outlier Detection : A Survey
A critical overview of outlier detection methods
Anomaly detection and critical attributes identification for products with multiple operating conditions based on isolation forest
Outlier Detection: Methods, Models, and Classification
Progress in Outlier Detection Techniques: A Survey
Recent Advances in Anomaly Detection Methods Applied to Aviation